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Device for fixing and positioning a cardiac pump

Abstract

The present document relates to a device for fixing a cardiac pump in an opening of a ventricular wall of a beating heart. The device includes: a hollow main body of overall cylindrical shape having an outer surface, this hollow main body includes a proximal end and a distal end between which said the outer surface extends, at least one portion of the outer surface of the main body intended to be placed inside ventricular cavity, except for its distal end, has a surface relief provided with protuberances and hollows, the distal end of the hollow main body forms a smooth crown having an arithmetic average roughness of less than or equal to 1 μm in order to stop the colonization of the fixing device by endothelial cells.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2011/0313237	12/2010	Miyakoshi	156/173	A61M 1/3653
2013/0261375	12/2012	Callaway	600/16	A61M 60/827
2015/0273124	12/2014	Callaway	623/3.26	A61M 60/857
2016/0067395	12/2015	Jimenez	606/151	A61M 60/178
2016/0175501	12/2015	Schuermann	600/16	A61M 60/17
2018/0256799	12/2017	Matthes	N/A	A61M 60/861
2021/0170164	12/2020	Wisniewski	N/A	A61M 60/861
2022/0395680	12/2021	Collas	N/A	A61M 60/865

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102018200974	12/2018	DE	A61F 2/2481
3103101	12/2020	FR	N/A
2010050114	12/2009	WO	N/A
WO-2012019126	12/2011	WO	A61M 1/12

OTHER PUBLICATIONS

Vrana, Nihal Engin et al. “Titanium microbead-based porous implants: bead size controls cell response and host integration.” Advanced healthcare materials vol. 3,1 (2014): 79-87. doi: 10.1002/adhm.201200369 (Year: 2014). cited by examiner
Han, Cheol-Min et al. “The electron beam deposition of titanium on polyetheretherketone (PEEK) and the resulting enhanced biological properties.” Biomaterials vol. 31,13 (2010): 3465-70. doi:10.1016/j.biomaterials.2009.12.030 (Year: 2010). cited by examiner
Mehdizadeh Omrani, Maryam et al. “Polyether ether ketone surface modification with plasma and

gelatin for enhancing cell attachment." Journal of biomedical materials research. Part B, Applied biomaterials vol. 109,5 (2021): 622-629. doi:10.1002/jbm.b.34726 (Year: 2020). cited by examiner

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based on and claims priority under 35 U.S.C. § 119 to French Patent Application No. 2111957, filed on Nov. 10, 2021, in the French Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.
BACKGROUND
Field
(2) The present document relates to the field of cardiac assist pumps.
(3) It relates more particularly to a medical device for fixing a cardiac pump in an opening of a ventricular wall of a beating heart.
Brief Description of Related Developments
(4) Cardiac insufficiency is a pathology in which a patient's heart is unable to deliver a sufficient flow of blood to meet the metabolic requirements of the organism.
(5) Cardiac assist pumps are conventionally used to assist a left ventricle of a heart. It is then referred to as an artificial heart pump. This artificial and mechanical pump does not replace the heart, which continues to function, but provides help to the weakened ventricle in order to increase the blood flow in a manner appropriate to the needs of the individual.
(6) In the case where a transplant is not possible, this cardiac pump is implanted on a long-term basis.
(7) As is illustrated in FIG. **1**, the implantable cardiac pumps of the prior art typically comprise an intraventricular part **2** and an extraventricular part **4**. In the extraventricular part **4**, a medical device with a cardiac pump of the prior art comprises from upstream (AM) to downstream (AV): an attachment flange **6**, a connector **8**, a removable sheath with non-return valve **10**, and a power cable **12**. In the intraventricular part **2**, this medical device has a cardiac pump **14** and an insert **16** made of polyether ether ketone (PEEK), said insert making it possible to receive, support and orient said cardiac pump **14** with respect to the aortic valve. The upstream and downstream parts are here located with respect to the ventricular wall of the beating heart.
(8) However, with a medical device of this kind, one observes a cellular colonization of the insert **16**, which extends to the part protruding from the body of the cardiac pump **14**. An obstruction of the blood aspiration orifices of the cardiac pump **14** thus appears over time, leading to poorer performance of the cardiac pump, which can have serious consequences as regards the health of the patient.
(9) There is therefore a pressing need for an insert with which it is possible to overcome the disadvantages of the prior art.
SUMMARY
(10) The present disclosure aims to overcome the disadvantages of the prior art by making available a device for fixing a cardiac pump in an opening of a ventricular wall of a beating heart, which device is simple in design and makes it possible to avoid obstruction of the blood aspiration

orifices of the cardiac pump, to maintain the latter in a desired orientation in the ventricular cavity, and to allow the ejection orifice of the cardiac pump to be positioned at a controlled distance from the aortic valve.
(11) The present disclosure also relates to a coating whose properties make it possible to facilitate a covering of tissue and to strengthen the anchoring of the main body of said fixing device.
(12) The present disclosure further relates to the use of a smooth crown which prevents any aggregation of proteins, cells or molecules, and which prevents colonization of the body of the pump by cell tissue within the ventricle.
(13) The present document relates to a device for fixing a cardiac pump in an opening of a ventricular wall of a beating heart, comprising a hollow main body of overall cylindrical shape having an outer surface, this hollow main body comprising a proximal end and a distal end between which said outer surface extends, said distal end being intended to protrude from said ventricular wall inside the corresponding ventricular cavity of the beating heart, at least one portion of the outer surface of said main body intended to be placed inside said ventricular cavity when the proximal end of this fixing device is fixed to said ventricular wall has, except for its distal end, a surface relief provided with protuberances and hollows and made of a material permitting the adherence and growth of endothelial cells, at least said portion of the outer surface comprising a coating covering a surface of titanium or of titanium alloy, the distal end of said hollow main body forms a smooth crown having an arithmetic average roughness R.sub.max of less than or equal to 1 μm in order to stop the colonization of said fixing device by endothelial cells.
(14) The smooth crown thus makes it possible to create a barrier to the colonization generating a space free of any natural tissue. The result of this is that the openings of the cardiac pump are no longer obstructed, and therefore the cardiac pump is no longer clogged up.
(15) Endothelialization of the coating of the main body is controlled by the surface state of said coating. Endothelialization is understood to mean a colonization by natural cell tissue. The coating of the main body promotes this endothelialization, by virtue of a surface relief provided with protuberances and hollows. This is promoted all the more when the coating is made of titanium or a titanium alloy. In fact, by this technique, the cellular adherence is improved.
(16) An advantage of this endothelialization is to guarantee anchoring and correct orientation of a system composed of main body and cardiac pump. In fact, this endothelialization makes it possible to exert a pressure on the system composed of main body and cardiac pump. This notion of orientation of the body of the cardiac pump is very important since it makes it possible not only to maintain the cardiac pump in place but also to arrange the latter properly and keep it stable with respect to the aortic valve. The pump is thus blocked at the desired depth in the ventricle.
(17) By virtue of this endothelialization, the coating of the main body is additionally preserved from any bacterial attack. The quantity of cell tissue agglomerating on the coating is optimized according to the geometry of the heart, whether this heart has an obtuse or oblique apex.
(18) Said hollow main body can have a first hollow cylindrical body made entirely of titanium or of titanium alloy, said first cylindrical body having, on at least part of its outer surface, said surface covering, a second hollow cylindrical body having an external flange at one end, said second cylindrical body being inserted into said first hollow cylindrical body such that its end is placed in the continuation of said outer surface of the first cylindrical body, forming a surface continuity therewith, said end of the second cylindrical body defining the distal end of said main body.
(19) This arrangement makes it possible not only to maintain the cardiac pump in place but also to arrange the latter properly and keep it stable with respect to the aortic valve.
(20) Said distal end or external flange can have a longitudinal dimension of between 10 mm and 20 mm.
(21) The quantity of cell tissue agglomerating on the coating is optimized according to the geometry of the heart. By virtue of this longitudinal dimension, the case of an obtuse heart is covered.

(22) The distal end or external flange can have a longitudinal dimension of between 2 mm et 10 mm.
(23) The quantity of cell tissue agglomerating on the coating is optimized according to the geometry of the heart. By virtue of this longitudinal dimension, the case of an oblique heart is covered.
(24) Said second cylindrical body can be smooth and made entirely of ceramic or of PEEK (polyether ether ketone).
(25) When the second body is made of PEEK, it is advantageously hydrophobic and inert. This second body is all the more hydrophobic and inert the more its surface is smooth. This second body does not support cell adherence.
(26) An arithmetic average roughness of the coating covering the outer surface made of titanium or of a titanium alloy can be between 100 μm and 300 μm.
(27) The coating of the main body promotes endothelialization, by virtue of the increased surface area of contact or more precisely by virtue of the high arithmetic average roughness.
(28) The proximal end of said main body can have a flared shape delimiting a seat for receiving a clamping ring, the function of which is to clamp annularly around the body of the cardiac pump, the inner wall of said proximal end moreover having a first, inner thread for screwing a toothed nut.
(29) This clamping ring permits leaktightness and holds the toothed nut in place.
(30) An outer wall of the proximal end can have a second, outer thread for receiving a ring comprising at least one lug, preferably four (4), each having an orifice for receiving the end of a clamping tool.
(31) Advantageously, this ring makes it easier for the operator to tighten the toothed nut. As the clamping tool comprises a recess intended to cooperate with the toothed nut, such as teeth complementing the teeth of the toothed nut, the free end of this clamping tool can be inserted into the opening of a lug in such a way that its recess is engaged with the teeth of the toothed nut for the purpose of tightening this nut.
(32) Furthermore, the distal end of said main body can be beveled in order to clamp the body of the cardiac pump inserted in said device, when said at least one part of the outer surface of the main body has been colonized by endothelial cells. The body of the pump is thus held firmly in position.
(33) The geometry of this beveled distal end makes it possible to adapt to the geometry of the beating heart and to permit better orientation of the cardiac pump.
(34) The coating can be formed solely of titanium microspheres.
(35) These titanium microspheres facilitate a tissue covering and solidify or strengthen the anchoring of the main body.
(36) The titanium microspheres can each have a mean diameter of between 100 μm and 300 μm.
(37) This diameter distribution is calculated to increase as far as possible the arithmetic average roughness of the coating.
(38) The coating can alternatively have an openworked woven fabric formed of a plurality of polyester filaments.
(39) The coating can be of the Spondycoat—T317A type.
(40) The outer surface of the coating can also have a scoured surface state. “Scoured surface state” is understood to mean a surface state for which a layer of material of the outer surface is removed, leaving a substrate exposed. For example, the state of this surface could be one that results from sandblasting the outer surface of the coating. During this sandblasting, an abrasive is sprayed at high speed, using compressed air, through a nozzle and onto the outer surface that is to be scoured.
(41) The coating can have protuberances and hollows in a random distribution.

Description

DESCRIPTION OF THE DRAWINGS

(1) Other features, details and advantages will emerge on reading the following detailed description and analyzing the appended drawings, in which:

(2) FIG. **1** shows a cardiac pump according to the prior art.

(3) FIG. **2** shows a first view of an assembly of a fixing device according to the disclosure.

(4) FIG. **3** shows a second view of a partially mounted assembly of the fixing device illustrated in FIG. **2**, according to the disclosure.

(5) FIG. **4** shows a view of a ring and a toothed nut of the fixing device, according to the disclosure.

(6) FIG. **5** shows a view, as per FIG. **4**, of the fixing device, showing a positioning of a clamping ring, according to the disclosure.

(7) FIG. **6** shows a downstream view of the fixing device with the toothed nut mounted in said fixing device, according to the disclosure.

(8) FIG. **7** shows a side view of the fixing device.

(9) FIG. **8** shows a sectional and schematic view of the fixing device.

(10) FIG. **9** shows the toothed nut fitted against the ring of the fixing device.

(11) FIG. **10** shows a first embodiment of a coating of a fixing tube, according to the disclosure.

(12) FIG. **11** shows a second embodiment of a coating of a fixing tube, according to the disclosure.

(13) FIG. **12** shows a third embodiment of a coating of a fixing tube, according to the disclosure.

(14) FIG. **13** shows a fourth embodiment of a coating of a fixing tube, according to the disclosure.

(15) FIG. **14** shows a cardiac pump according to the disclosure.

DETAILED DESCRIPTION

(16) The drawings and description below essentially contain elements of a certain character. Therefore, they not only may be used to better understand the present disclosure, but also contribute to its definition, where applicable. It will be noted that the figures are not to scale.

(17) The present document relates to a device **18** for fixing a cardiac pump in an opening of a ventricular wall of a beating heart.

(18) As is illustrated in FIGS. **2** to **8**, the fixing device **18** has a hollow main body **20** of overall cylindrical shape. This main body **20** comprises a distal end **22** and a proximal end **24**. Distal end **22** is understood as the end of the hollow main body **20** farthest away from the ventricular wall of the beating heart. Conversely, proximal end **24** is understood as the end of the hollow main body **20** that is closest to the ventricular wall of the beating heart. The hollow main body **20** comprises a first hollow cylindrical body **26** and a second hollow cylindrical body **28**. The first hollow cylindrical body **26** is made of titanium or of a titanium alloy. The second hollow cylindrical body **28** is made entirely of ceramic or of PEEK (polyether ether ketone).

(19) The first cylindrical body **26** of the main body **20** comprises, at the proximal end **24**, a flared shape delimiting an inner seat. The flared shape can be conical, for example. The seat delimits a space inside the first cylindrical body **26** of the main body **20** able to receive a clamping ring **30**. This clamping ring **30** is able to deform, so as to conform to an inner wall of the first cylindrical body **26** of the main body **20** in proximity to said proximal end **24**.

(20) At the proximal end **24**, the first cylindrical body **26** has a first, inner thread **32** and a second, outer thread **34**. The first, inner thread **32** of the first cylindrical body **26** is configured to receive a toothed nut **36** that can be screwed into said first, inner thread **32** and come into contact with an end **38** of the clamping ring **30**. The second, outer thread **34** of the first cylindrical body **26** is configured to receive a ring **40**. This ring **40** can be screwed along said second, outer thread **34**. The ring **40** has four lugs **42**. These lugs **42** each have a receiving orifice **44** for a clamping tool **46**. As is illustrated in FIG. **89**, the clamping tool **46** has a free end **48** that cooperates with the toothed nut **36**.

(21) An outer surface **52** of the first cylindrical body **26**, excluding the distal end **22** of this first

cylindrical body **26**, has a coating **54**. This coating **54** has a surface relief provided with a random distribution of protuberances and hollows. This coating, covering the outer surface **52** of the first cylindrical body **26** except for the distal end **22**, has a parameter of arithmetic average roughness of between 100 μm and 300 μm.

(22) In a particular embodiment illustrated in FIG. **10**, this coating **54** can comprise a plurality of layers of microspheres **56** of titanium. These microspheres **56** of titanium have a mean diameter of between 100 μm and 300 μm. This coating **54** has a surface relief characterized by an arithmetic average roughness of between 100 and 300 μm. These microspheres **56** are sprayed onto the outer surface **52** of the first cylindrical body **26**, excluding the distal end. The microspheres **56** are bound to the outer surface **52** by heating to a temperature close to the melting point of titanium. No binder is used to hold the microspheres **56** together.

(23) In a particular embodiment illustrated in FIG. **11**, this coating **54** can comprise an openworked woven fabric **58** formed of a plurality of polyester filaments. The filaments have a mean diameter of between 250 μm and 350 μm. This woven fabric has openings **60** with a mean diameter of between 50 μm and 100 μm.

(24) In a particular embodiment illustrated in FIG. **12**, the coating **54** can have a scoured surface state **62**. The result of this scouring is that a granularity is present at the surface, increasing the surface area of contact between said coating and the blood circulating in the left ventricle of the heart. This granularity can be quantified in terms of arithmetic average roughness. The arithmetic average roughness of said coating is between 100 and 300 μm. This scouring is obtained by sandblasting.

(25) In a particular embodiment illustrated in FIG. **13**, the outer surface has received a surface treatment. The coating is composed of PEEK that has received a plasma spray. Traditionally, plasma is a partially ionized gas composed of atoms, molecules, ions and excited free radicals, following stimulation by radio frequencies, microwaves or electron discharge. This plasma spray is configured to influence a hydrophilic/hydrophobic character, a charge and a surface roughness. The coating can be of the Spondycoat 64—T317A type.

(26) The second cylindrical body **28** is able to be inserted inside the first hollow cylindrical body **26** such that a first part **66** is in contact against an inner surface of the second hollow cylindrical body **26** and a second part **68** protrudes from a distal end **70** of the first hollow cylindrical body **26**, this distal end **70** of the first cylindrical body **26** being opposite the second, outer thread **34**. This second part **68** of the second cylindrical body **28** forms the distal end **22** of the main body **20**. This second part **68** has an external flange **72**. This external flange **72** comprises a smooth crown **74**. This smooth crown **74** has an arithmetic average roughness Rmax of less than or equal to 1 μm in order to stop the colonization of said fixing device **18** by endothelial cells. The distal end **22** of the main body **20**, hence the smooth crown **74**, is beveled. This beveled smooth crown **74** is configured to clamp the cardiac pump **76**. As is illustrated in FIG. **14**, the cardiac pump **76** is inserted from the side of the toothed nut **36**, passes inside the main body **20** and emerges from the side of the smooth crown **74**.

(27) A longitudinal dimension by which the smooth crown **74** extends depends on a span of the heart and on a thickness of a wall of said heart. Longitudinal dimension is understood as a space between the distal end **70** of the first cylindrical body **26** and a distal end **76** of the smooth crown **74**. It can also be referred to as a depth of the smooth crown **74**.

(28) The depth of the smooth crown is between 2 and 10 mm, if the heart has a very obtuse apex at the end of contraction. In fact, during contraction, there is then very little contact between the smooth crown and the wall of the heart.

(29) By contrast, the depth of the smooth crown **74** is between 10 et and 20 mm if the heart has a very obtuse apex at the end of contraction. In fact, the increase in the depth of the smooth crown **74** is configured to avoid a situation where the walls of the heart are not in contact with this smooth crown **74**. There is therefore no deposition of cells on said smooth crown **74**.

(30) During operation, the smooth crown **74** makes it possible to create a barrier to the colonization generating a space free of any natural tissue. In fact, the smooth crown **74** made of PEEK is hydrophobic and inert. This smooth crown **74** is all the more hydrophobic and inert the more its surface is polished; the distal end of the smooth crown **74** will not support cellular adherence. The result of this is that openings **78** of the cardiac pump **76** are no longer obstructed, and therefore the cardiac pump **76** is no longer clogged up.
(31) Endothelialization of the coating **54** of the first cylindrical body **26** is controlled by the surface state of said coating **54**. Endothelialization is understood to mean a colonization by natural cell tissue. The coating **54** of the first cylindrical body **26** promotes endothelialization by virtue of the increased surface area of contact or more precisely by virtue of the high arithmetic average roughness. In fact, by this technique, the cellular adherence is improved. The advantage of this endothelialization is to reinforce the correct orientation of a system composed of main body **20** and cardiac pump **76**. In fact, this endothelialization makes it possible to exert a pressure on the system composed of main body **20** and cardiac pump **76**. This notion of orientation of the body of the cardiac pump **76** is very important since it makes it possible not only to maintain the cardiac pump **76** in place but also to arrange the latter properly and keep it stable with respect to an aortic valve. By virtue of this endothelialization, the coating **54** of the first cylindrical body **26** is preserved from any bacterial attack. The quantity of cell tissue agglomerating on the coating **54** is optimized according to the geometry of the heart, whether this heart has an obtuse or oblique apex.
(32) The use of the second cylindrical body **28** makes it possible to avoid scratching the cardiac pump **76** while using a softer material.

Claims

1. A device, for fixing a cardiac pump in an opening of a ventricular wall of a beating heart, comprising: a hollow main body of overall cylindrical shape having an outer surface, this hollow main body comprising a proximal end and a distal end between which said outer surface extends, said distal end being intended to protrude from said ventricular wall inside the corresponding ventricular cavity of the beating heart, at least one portion of the outer surface of said main body intended to be placed inside said ventricular cavity, except for its distal end, has a surface relief provided with protuberances and hollows and made of a material permitting the adherence and growth of endothelial cells, at least said portion of the outer surface comprising a coating covering a surface of titanium or of titanium alloy, and the distal end of said hollow main body forms a smooth crown having an arithmetic average roughness of less than or equal to 1 μm in order to stop the colonization of said fixing device by endothelial cells, wherein said hollow main body has: a first hollow cylindrical body made entirely of titanium or of titanium alloy, wherein said first cylindrical body has, on at least part of its outer surface, said surface coating, a second hollow cylindrical body having an external flange at its end, wherein said second cylindrical body is inserted into said first hollow cylindrical body such that its end is placed in the continuation of said outer surface of the first cylindrical body, forming a surface continuity therewith, and wherein: the outer surface of the portion of the external flange in contact with the first cylindrical body and the outer surface of the portion of the first cylindrical body in contact with the external flange is flush, the rest of the outer surface of said external flange has a diameter equal to and/or less than the diameter of the outer surface of the first cylindrical body, said end of the second cylindrical body defines the distal end of said main body, an opening defined by the hollow main body being configured to receive a propulsive cardiac pump body, said end of the second cylindrical body being configured to maintain in position said propulsive cardiac pump body when inserted; and wherein said second cylindrical body is made of PEEK (polyether ether ketone) and has an entirely smooth surface with an arithmetic mean roughness Rmax of less than or equal to 1 μm, so as to prevent said second cylinder body from scratching of the propulsive cardiac pump.

2. The device according to claim 1, in which said distal end or external flange has a longitudinal dimension of between 10 mm and 20 mm.
3. The device according to claim 1, in which the distal end or external flange has a longitudinal dimension of between 2 mm and 10 mm.
4. The device according to claim 1, in which an arithmetic average roughness of the coating covering the outer surface made of titanium or of a titanium alloy is between 100 μm and 300 μm.
5. The device according to claim 1, in which the proximal end of said main body has a flared shape delimiting a seat for receiving a clamping ring to clamp annularly around the body of the cardiac pump, the inner wall of said proximal end having a first, inner thread for screwing a toothed nut.
6. The device according to claim 5, in which an outer wall of the proximal end has a second, outer thread for receiving a ring comprising at least one lug having an orifice for receiving the end of a clamping tool.
7. The device according to claim 1, in which the distal end of said main body is beveled in order to clamp the body of the cardiac pump inserted in said device.
8. The device according to claim 1, in which the coating is formed solely of titanium microspheres.
9. The device according to claim 8, in which the titanium microspheres each have a mean diameter of between 100 μm and 300 μm.
10. The device according to claim 1, in which the coating has an open worked woven fabric formed of a plurality of polyester filaments.
11. The device according to claim 1, in which the coating is a plasma spray.
12. The device according to claim 1, in which the outer surface of the coating has a scoured surface state.
13. The device according to claim 1, in which the coating has protuberances and hollows in a random distribution.